UHF PA Design Exercise

July 22, 2026

This document is a record of some experiments with a two stage PA board using a generic gain block as a driver, and RDMUS01 mosfet final stage, conducted in July 2026 by Tibor, with David taking notes. Some details of the module including a complete schematic are presented on the KA7OEI blog. The aim was to develop a stable, 1W PA for 400-450 MHz (including the Ham 70 cm band). The general approach was:

- Start with the generic unmatched, broadband module (untuned 50 Ω matches throughout).
- Add a $R_{deg} = 20$ Ω degeneration resistor across the mosfet gate to ensure stability (and the cost of some gain). Reducing this to 10 ohms resulted in insufficient drive to the final stage.
- Build an interstage match to transform the 50 ohm driver output impedance to 20 Ω mosfet gate impedance.
- We chose $V_{dd} = 7.6$ V such that a 2x swing on the mosfet output will be well less than $V_{vddmax} = 25$ V.
- We chose an output impedance for the mosfet of 20 Ω to obtain 1W.
- We designed an output match for 20 to 50 Ω. This was initially a L network, we later experimented with a π network, which had better harmonic suppression.

General Tips:

- Use vaseline as a non-toxic thermal grease between mosfet and heatsink.
- Use physically large, high $Q \approx 100$ commercial inductors for matching. Not hand wound air core or wire loops, as they are excellent antennas.
- Use 100 pF bypass capacitors rather than 1 nF, as lower impedance.
- Matching C in output stage was adjusted to achieve flat gain between $400 - 450$ MHz.

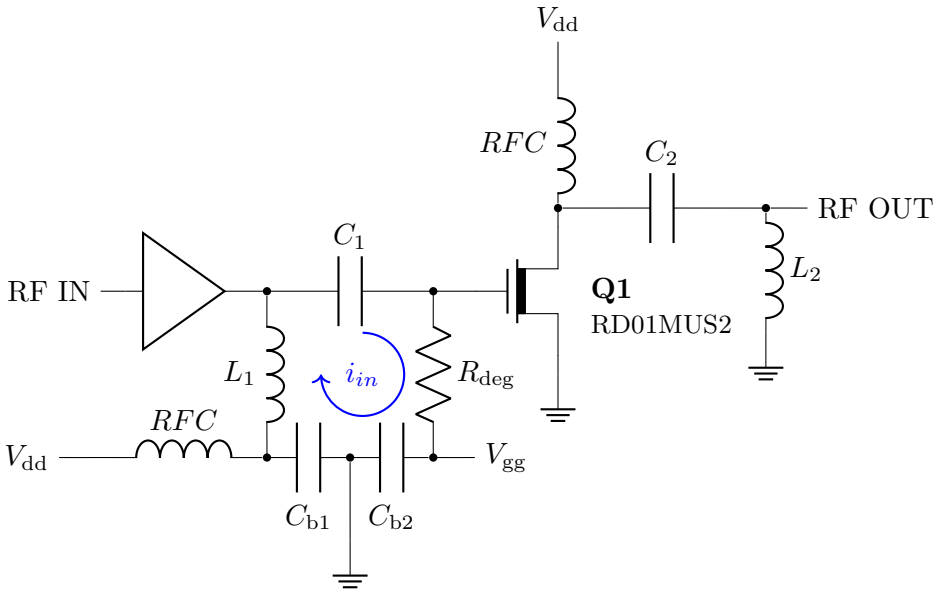


Figure 1: Two stage prototype PA. An input match is formed from C_1 and L_1, and an output match by C_2 and L_2, both with high pass topologies. R_{deg} is the degeneration resistance that ensures stability. V_{gg} is the DC gate bias. C_{b1} and C_{b2} are DC blocks. The layout should make the current loop i_{in} as small as possible.

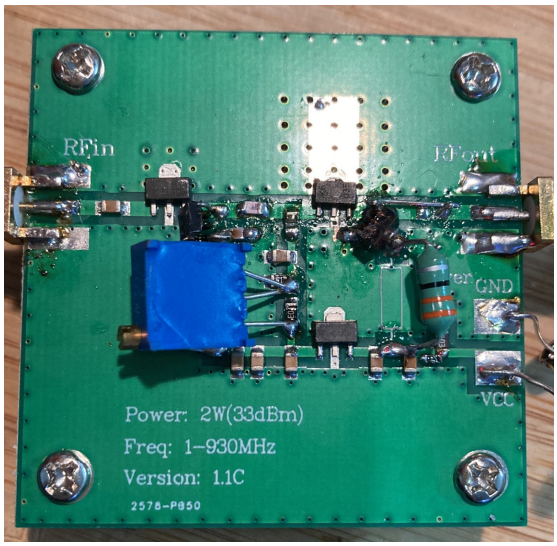


Figure 2: Two stage PA PCB used for prototyping, modified to have an interstage and final stage high pass matches. Trimmer provides V_{gg} gate bias adjustment.

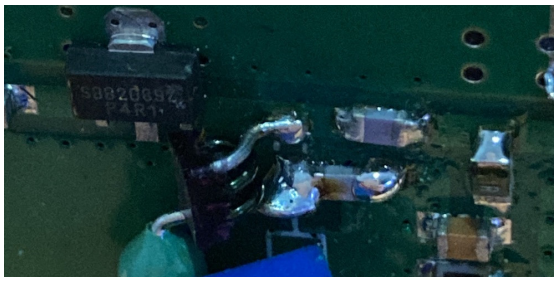


Figure 3: Interstage match layout: C_{b1} (lower middle) and C_{b2} (lower right) ground nodes connected close together to minimise current loop i_{in}. R_{deg} is on RHS, C_1 in middle.

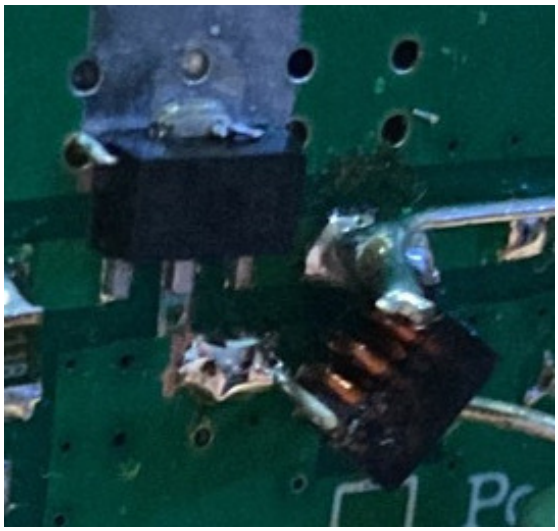


Figure 4: Output shunt matching component (in this iteration an inductor) is connected directly to the mosfet source pin to minimise the current loop. A better ground connection was later found to be the plane area (top), which is connected to the pad on the bottom of the mosfet.

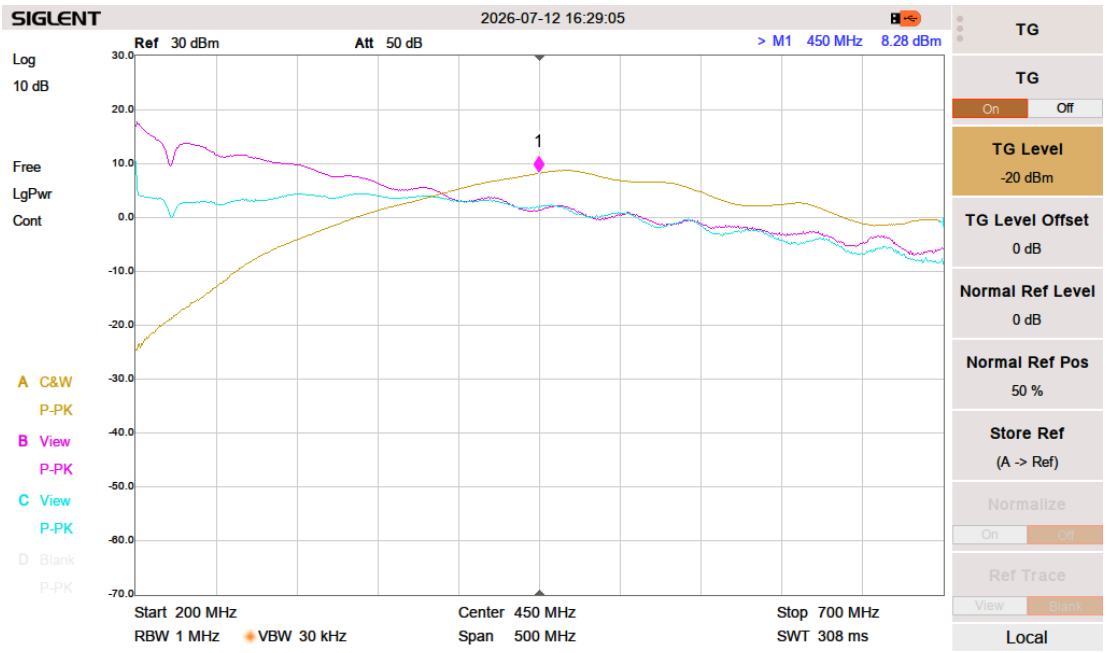


Figure 5: Purple unmodified original, no matches, straight through 50 ohm design. Blue is with gate degeneration 10 ohm resistor, but no matching. Yellow is the completed first prototype with high pass interstage match and high pass output match.

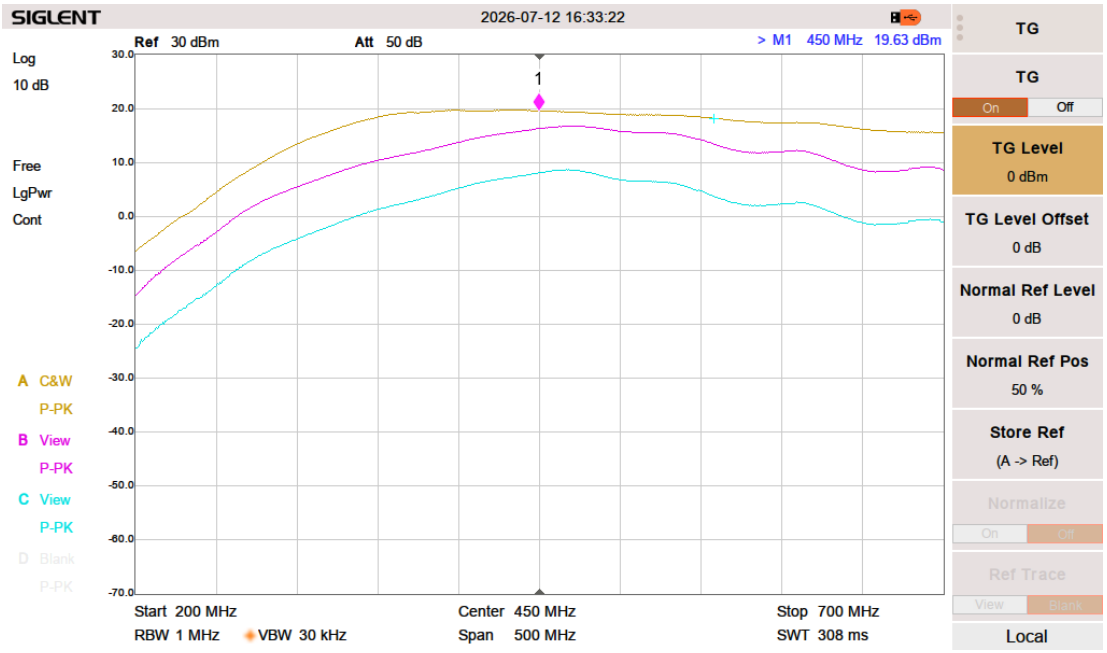


Figure 6: Performance across drive levels. Blue -20 dBm, pink -10 dBm, yellow 0 dBm drive (compressed). When compressed, power gain is 30 dB, when uncompressed 36 dB.